

High-burnup structure formation as a second-order phase transition

A Landau functional for SCIANTIX

iHighBurnupStructureFormation = 4

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SCIANTIX status

The model

Results

Model applied to SCIANTIX

SCIANTIX status

SCIANTIX had three formation models for the restructured volume fraction

option	form	driven by
1	KJMA, Barani (2020) ¹	effective burnup
2	KJMA + $bu_{inc} = 15$, Biswas & Aagesen (2025) ²	effective burnup
3	KJMA on ρ_d , Veshchunov & Shestak (2009) ³	local burnup

All three are **transformation kinetics**: a sigmoid fitted in burnup. None of them produces the **mean misorientation** Θ of the subgrains, or the **subgrain radius** r_n .

¹ Barani et al., *J. Nucl. Mater.* 539 (2020) 152296.

² Biswas & Aagesen, *Comput. Mater. Sci.* 258 (2025) 114052, Eq. 45.

³ Veshchunov & Shestak, *J. Nucl. Mater.* 384 (2009) 12–18.

The data this model is built on

EBSD — Zacharie-Aubrun *et al.* (2022), Onofri *et al.* (2025). 42 radial positions on 7 rods, with TRANSURANUS local conditions.

quantity	N	what it is
mean misorientation Θ	41	$[A_{\text{Mis}}(f_1 - f_{10}) + 10f_{10}]/100$
restructured fraction X	27	f_{10} , area fraction above 10°
subgrain radius r_n	14	ECD50 % /2

f_1 is the same area fraction taken at 1° instead of 10° . That 1° is the lower bin edge of the whole dataset — below it a boundary is not counted — and it is what fixes θ_u later, in Eq. (10).

What KJMA is, and what it assumes

Kolmogorov, Johnson & Mehl and Avrami, independently, 1937–41.⁴ Nuclei appear *at random* in the untransformed material and grow *isotropically*; count the transformed volume as if the growing regions could overlap freely — the **extended volume** V_e — and the overlap correction is purely statistical:

$$dX = (1 - X) \frac{dV_e}{V} \implies X = 1 - \exp(-K t^n)$$

What you get

Two numbers, K and n , and a sigmoid. n carries the dimensionality of growth and the nucleation regime — but only if the assumptions hold.

What it assumes

Randomly placed nuclei, a growth law independent of where you are in the transformation, and a *clock*. In SCIANTIX the clock is burnup.

⁴ Kolmogorov (1937); Johnson & Mehl, *Trans. AIME* 135 (1939) 416; Avrami, *J. Chem. Phys.* 7 (1939) 1103. Derivation: Christian (2002), ch. 12.

The three options are three answers to “which clock?”

- 1 **Effective burnup**, $n = 3.54$, $K = 2.77 \times 10^{-7}$, fitted on restructured-fraction data. The transformation begins the instant irradiation does.
- 2 Same, with an **incubation burnup** $bu_{inc} = 15 \text{ MWd/kgHM}$ below which nothing happens. Biswas & Aagesen get it from a balance between the energy stored in dislocations and the energy of forming a subgrain — **the same physical argument as this work**, used to shift the origin of an empirical curve rather than to build the curve.
- 3 KJMA not on burnup but on the **dislocation density** $\rho_d(bu, T)$, with a threshold $\rho_{crit} = 6 \times 10^{14} \text{ m}^{-2}$. The first to make the driver a **microstructural state variable** instead of a clock, and the only one of the three that feels temperature.

What none of them can do

They are calibrated on X alone. There is no Θ , no r_n , and no internal state: ask the model where the microstructure *is* and it can only tell you how far along a fitted curve it has travelled.

Why a phase transition is the right shape for this problem

The HBS forms *continuously*

Not by nucleating new grains, but by low-angle boundaries absorbing dislocations until their misorientation drifts through the 10° HAGB threshold. In the metallurgical literature this is *continuous* dynamic recrystallization, and it is a well-posed alternative to the nucleation-and-growth picture KJMA describes.⁵

The evidence in UO_2 is the EBSD itself: Θ rises smoothly with burnup. There is no population of already- 10° nuclei waiting to grow.

A continuous transformation has an order parameter

If a single continuous quantity distinguishes the two states, Landau's argument applies: expand the free energy in it, and keep only the terms symmetry allows.⁶

$$F(\eta) = C_0 + C_2\eta^2 + C_4\eta^4$$

Odd powers are forbidden here for a concrete reason — the energy of a boundary cannot depend on the *sign* of the misorientation across it.

C_2 changing sign *is* the transition. Nothing is fitted to make it happen.

⁵Gourdet & Montheillet, *Acta Mater.* 51 (2003) 2685; Humphreys & Hatherly (2017), chs. 6 and 10.

⁶Landau (1937); Landau & Lifshitz, *Statistical Physics I*, §143.

Landau against KJMA, side by side

	KJMA (options 1–3)	Landau (option 4)
what is solved	a <i>kinetic</i> law along a path	the <i>equilibrium</i> of a state
driver	a clock (burnup), or ρ_d	$\rho_{\text{tot}}(bu)$ through C_2
free parameters	K , n fitted on X	β , k , ρ_c fitted on Θ , r_n
predicts X	by construction	as a by-product
predicts Θ , r_n	no	yes
reversible?	no, monotone by construction	yes — see below

The price of using an equilibrium theory

F is minimised afresh at every step, so a drop in ρ_{tot} would un-restructure the fuel.

Restructuring is irreversible, so the implementation carries a **monotonic lock** on Θ , and re-derives r_n and X from the locked value. It is the one piece of the model that is kinetics rather than thermodynamics.

The model

HBS as a second-order phase transition

Order parameter

$$\eta = \frac{\theta}{\theta_{\max}}, \quad \theta_{\max} \equiv \theta_{\text{HAGB}} = 10^\circ$$

the mean misorientation normalised to the LAGB/HAGB boundary.

External condition: the local burnup, entering *only* through the dislocation density

$$\rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

Nogita & Une (1994)

Free energy

$$F(bu, \eta) = C_0 + C_2 \eta^2 + C_4 \eta^4$$

Only even powers: the energy cannot depend on the *sign* of a misorientation.

Minimising F gives Θ ; the same wall geometry gives r_n ; the lever rule gives X .

Three adjustable parameters

β (wall geometry), k (sweeping), ρ_c (strain-field cut-off).

The geometric wall

A low-angle boundary *is* a wall of dislocations. Two classical relations fix the whole geometry.

1. Spacing sets the angle $\theta = b/d$

so a wall of n dislocation families carries a line length per unit *wall area*

$$L_A = n\theta/b$$

2. Wall area per unit volume

$$S/V = 3\sqrt{\rho_{\text{LAGB}}} / \beta$$

β : the one geometric parameter (Rest & Hofman 2000).
 $n = 2$, Gourdet & Montheillet, not fitted (between 1-3).

Close the two on each other:

$$\rho_{\text{LAGB}} = L_A \cdot S/V$$

$$\Rightarrow \sqrt{\rho_{\text{LAGB}}} = \frac{3n\theta}{\beta b}$$

Everything follows from θ alone

$$\rho_{\text{LAGB}} = \left(\frac{3n\theta}{\beta b} \right)^2, \quad r_n \simeq \frac{\beta^2 b}{6n\theta}$$

The dislocation balance

Every dislocation is in exactly one of three states, and they must add up to ρ_{tot} :

$$\underbrace{\rho_{\text{ord}} = \rho_{\text{LAGB}}^{\text{max}} \eta^2}_{\text{condensed into low-angle walls}} + \underbrace{\rho_{\text{swept}} = (\rho_{\text{tot}} - \rho_{\text{ord}}) \frac{\Delta R}{R}}_{\text{annihilated by moving boundaries}} + \underbrace{\rho_{\text{free}}}_{\text{still a random array}} = \rho_{\text{tot}}$$

Each population carries the line energy of the state it is in:

$$F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2, \quad A_i = \frac{f(\nu)}{4\pi} \ln \frac{R_i}{b}$$

free ones screened at ρ_c , wall ones by the wall itself at ρ_{tot} ; the swept ones carry nothing.

The functional, collected

$$C_0 = \rho_{\text{tot}} A_1 G b^2, \quad C_4 = \rho_{\text{LAGB}}^{\text{max}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2$$
$$C_2 = \underbrace{\rho_{\text{LAGB}}^{\text{max}} (A_2 - A_1) G b^2}_{\text{walls: the screening gain}} - \underbrace{\rho_{\text{tot}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2}_{\text{sweep}}$$

The engine of the transition is $A_2 - A_1$. As burnup rises, ρ_{tot} rises, the screening length inside the wall shrinks, A_2 falls below A_1 — and it becomes cheaper to store dislocations in walls than to leave them random.

Equilibrium

$$\frac{\partial F}{\partial \eta} = 0 \implies \eta^2 = -\frac{C_2}{2C_4}, \quad \text{clipped at 0}$$

$$\eta = 0 \text{ wherever } C_2 \geq 0.$$

Condition on the minimisation problem: dislocation balance

The walls cannot hold more dislocations than exist, so $\rho_{\text{ord}} \leq \rho_{\text{tot}}$. The equilibrium is the minimum of F on the **admissible interval**, not the free stationary point:

Eq. (7b) — admissibility

$$\eta_{\text{bal}} = \sqrt{\min\left(\frac{\rho_{\text{tot}}}{\rho_{\text{LAGB}}^{\text{max}}}, 1\right)}, \quad \eta_{\text{eq}} = \min\left(\sqrt{-\frac{C_2}{2C_4}}, \eta_{\text{bal}}\right)$$

Results

Calibration

Joint fit of β, k, ρ_c on Θ and the subgrain sizes, while the restructured fraction is *not* in the objective.

$$J = \frac{\langle \Delta \Theta^2 \rangle}{\text{var } \Theta} + w \frac{\langle \Delta r^2 \rangle}{\text{var } r}$$

dimensionless and symmetric;
differential_evolution, 6 seeds, all
converging to $J = 0.2398$.

n (not fitted)	2
β	33.547
k	0.04697
ρ_c	$1.166\text{e}15 \text{ m}^{-2}$

Three parameters, two datasets, three predictions

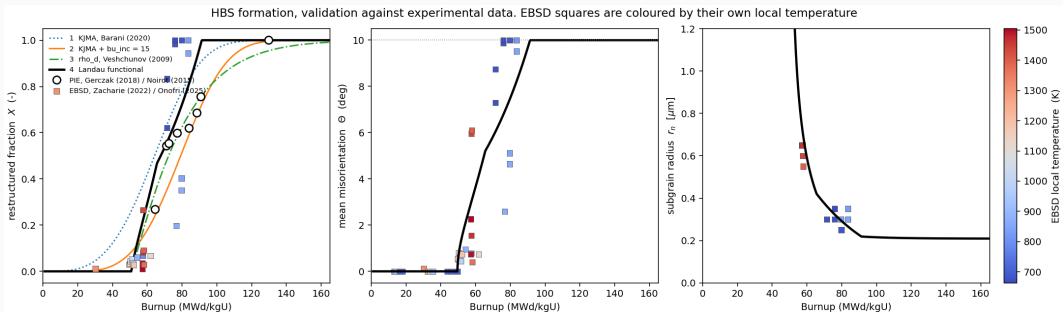
β, k and ρ_c are fitted on Θ and r_n only. The restructured fraction X — the one quantity SCIANTIX actually consumes — is left **out of the objective** and comes out of the lever rule, Eq. (10), as a **prediction**.

That is what makes the next slide a test rather than a report of the fit: X and the PIE fractions are scored on data the calibration never saw.

What the balance bought

		RMSE / R^2
Θ	($N = 41$, calibrated)	1.7616° / 0.772
X	($N = 27$, <i>not</i> fitted)	0.2122 / 0.710
r_n	($N = 14$, calibrated)	$0.0678 \mu\text{m}$ / 0.774
PIE fraction ($N = 8$, out-of-sample)		0.1461 / +0.433

Validation against both datasets



Against the three models SCIANTIX already had

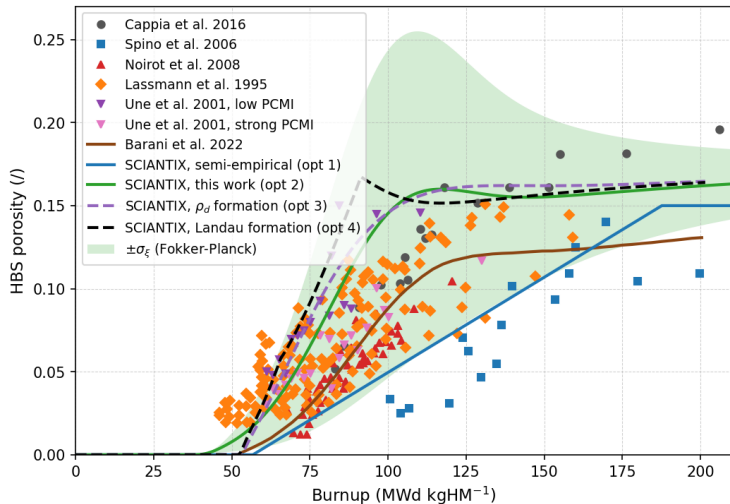
Same history, same initial conditions, same time stepping; only the formation model varies.
Burnup [MWd/kgU] at which α_r first reaches:

threshold	1 KJMA	2 KJMA+ bu_{inc}	3 ρ_d	4 Landau
1 %	19.5	34.5	51.2	51.1
10 %	37.8	52.8	54.8	53.5
50 %	64.2	79.3	72.3	67.7
90 %	90.1	105.2	112.3	87.7
99 %	109.7	124.6	161.7	91.1

- At the onset, option 4 lands **within 0.1 MWd/kgU of option 3** — an independently calibrated model built on a different microstructural variable, so a genuine cross-check and not a shared fit.
- They part at the top: the lever rule **saturates** when Θ hits the 10° cap, whereas KJMA approaches 1 asymptotically. Option 4 is done by 91 MWd/kgU, option 3 not before 162.
- Option 4 spans the transition in **40 MWd/kgU** against 90 for option 1: once the driver is 16/17 a state variable rather than a fitted sigmoid in burnup, the transformation is sharper

Model applied to SCIANTIX

Does the new formation model disturb the porosity model?



Largest deviation **+0.044**
absolute at
91.3 MWd/kgHM
(+36 %), exactly the
saturation burnup.
Elsewhere: ≤ 0.026
below 75, ≤ 0.008 above
120, and the two end
states are **0.7 % apart**.

Held against option 2 in
test_U02HBS; all four
options end within **0.17 %**
of the same porosity.

Summary

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- [4] M. Avrami, *J. Chem. Phys.* **7** (1939) 1103–1112; **8** (1940) 212–224; **9** (1941) 177–184.
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- [8] S. Gourdet, F. Montheillet, *A model of continuous dynamic recrystallization*, *Acta Mater.* **51** (2003) 2685–2699. *n*, and the sweeping term
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- [14] T. Barani *et al.*, *Modeling high burnup structure in oxide fuels. . . Part I: HBS formation*, *J. Nucl. Mater.* **539** (2020) 152296. *option 1*
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- [17] M.S. Veshchunov, V.E. Shestak, *Model for evolution of crystal defects in UO_2 under irradiation up to high burn-ups*, *J. Nucl. Mater.* **384** (2009) 12–18. *option 3, $\rho_d(\text{bu}, T)$*
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28 of the 42 EBSD points
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the other 14
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Backup — the equations

$$(1) \quad \rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

$$(2) \quad G(T, P, x, q), \quad \nu(T, P, x, q) \quad \text{NEA/NSC/R(2024)1 p. 124;} \quad f(\nu) = \frac{1-\nu/2}{1-\nu}$$

$$(3) \quad \rho_{\text{LAGB}}^{\max} = \left(\frac{3n\theta_{\max}}{\beta b} \right)^2, \quad (S/V)_{\max} = \frac{9n\theta_{\max}}{\beta^2 b}, \quad \left. \frac{\Delta R}{R} \right|_{\max} = \frac{k \rho_{\text{LAGB}}^{\max}}{\rho_{\text{tot}}}$$

$$(4) \quad A_1 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_c^{-1/2}}{b}, \quad A_2 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{tot}}^{-1/2}}{b}$$

$$(5,6) \quad F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2 = C_0 + C_2 \eta^2 + C_4 \eta^4$$

$$(7,7b) \quad \eta_{\text{eq}} = \min(\sqrt{-C_2/2C_4}, \eta_{\text{bal}}), \quad \eta_{\text{bal}} = \sqrt{\min(\rho_{\text{tot}}/\rho_{\text{LAGB}}^{\max}, 1)}$$

$$(8) \quad \Theta = \min(\eta_{\text{eq}} \theta_{\max} \cdot \frac{180}{\pi}, \theta_{\text{HAGB}})$$

$$(9) \quad r_n = \min\left(\frac{1.5}{(S/V)}\left(1 + \frac{\Delta R}{R}\right), R_{\text{grain}}\right)$$

$$(10) \quad X = \text{clip}\left(\frac{\Theta - \theta_u}{\theta_{\text{HAGB}} - \theta_u}, 0, 1 - 10^{-9}\right)$$

Backup — regimes and parameters

Regimes (T -, P -, x -independent)	
condition	bu [GWd/tU]
$C_2 = 0$, Θ leaves 0	49.56
$\Theta = \theta_u$, X leaves 0	50.81
$\Theta = \theta_{\text{HAGB}}$	91.32

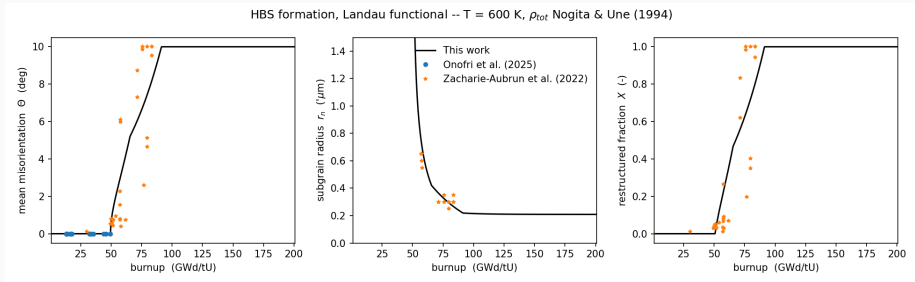
The balance bound sets η over roughly **65.6–91.3 GWd/tU**, and at 11 of the 41 EBSD points.

Fixed constants

b	3.889e-10 m
θ_{max}	$= \theta_{\text{HAGB}}$
θ_{HAGB}	10°
θ_u	1.00°
n	2

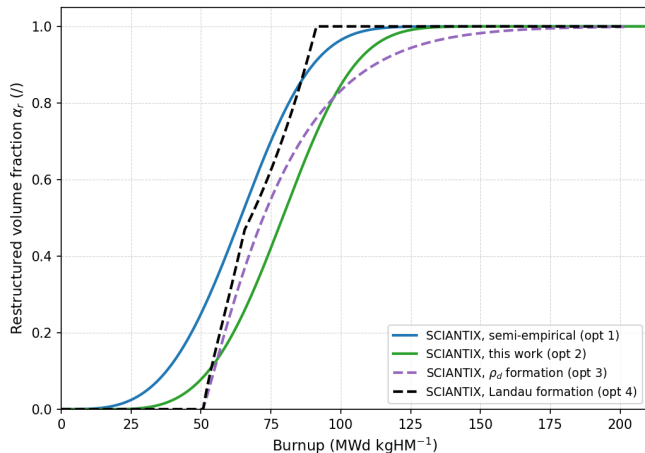
θ_u is the lower end member of the mixture Eq. (10) inverts, and it is set by the **measurement convention**, not fitted: the EBSD data reports a restructured fraction *at* 1° (f_1) and one *at* 10° (f_{10}), so 1° is the edge below which a boundary is not counted at all. A map whose whole resolved population sits there carries no restructuring — which is $X = 0$. It fixes only *where* X leaves zero: Θ and r_n do not depend on it.

Backup — the model on its own, $T = 600$ K, $P = 0.05$



```
python3 hbs_formation_landau.py --plot
```

Backup — the four regression cases, as they ship



Not the controlled comparison of the previous figure: each case pairs its formation option with the porosity model it was built around (1, 2, 3, 3), so part of the spread is the porosity model.

This is what a user of each regression case actually gets;
`formation_options.png` is what isolates the formation model.

Backup — the four formation options in a full run

HBS formation options on the same irradiation, porosity model 3 throughout

